

Rounding the (Non)Bayesian Curve: Unraveling the Effects of Rounding Errors in Belief Updating

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Introduction

- ▶ Beliefs are fundamental to economic models of decision-making
 - ▶ Ex: households' inflation expectations, investors' response to unexpected earnings, workers' belief about likelihood of receiving job offers based on economic conditions
- ▶ Deviations from Bayesian updating have received a lot of attention in behavioral and experimental literature
 - ▶ **Seminal**: Phillips and Edwards (1966); Edwards (1968); Tversky and Kahneman (1974); El-Gamal and Grether (1995);
 - ▶ **More Recent**: Holt and Smith (2009); Moreno and Rosokha (2016); Charness and Dave (2017); Hill (2017); Ambuehl and Li (2018); Coutts (2019); Benjamin (2019); Bland and Rosokha (2021); Oprea and Yuksel (2023); Mobius, Niederle, Niehaus, Rosenblat (2022), Aydogan, Baillon, Kemmel, Li (2023); Fan, Liang, Peng (2023); Esponda, Vespa, Yuksel (2024); Augenblick, Lazarus, and Thaler (2025).

Various Approaches

▶ Elicitation

- ▶ Becker-DeGroot-Marschak (BDM)
- ▶ Quadratic Scoring Rule (QSR)
- ▶ Binirized Scoring Rule (BSR)
- ▶ Probability Matching
- ▶ Exchangeable Events
- ▶ Multiple Price List (in \$ space)

▶ Task

- ▶ Neutral vs Ability
- ▶ Inference vs Forecast

▶ Estimation

- ▶ Reduced-form
- ▶ Structural

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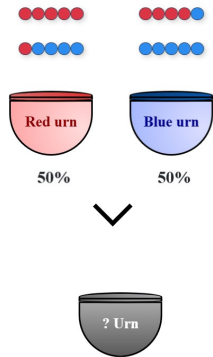
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- ▶ Ignored issues (until now): **rounding and heterogeneity**
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Typical Scenario



- ▶ Symmetric with respect to the composition of the urns (*i.e.*, *signal strength*)
- ▶ Round priors (*typically 50/50*)
- ▶ All subjects face the same set of scenarios (*draws may vary*)
- ▶ Elicitations equally spaced across signals (*BSR question after each of the first three draws*)

History of Draws:



What We Do

- ▶ Meta-study of recent papers to document rounding
 - ▶ Connect rounding to estimates of conservatism
- ▶ Simulations to explore issues with the common estimation approach
 - ▶ Rounding, Heterogeneity bias
- ▶ Experiment to understand impact of the issues on the estimates
 - ▶ Novel features
- ▶ Propose an econometric solution
 - ▶ Interval-valued approach + hierarchical model estimation

Background

Common Reduced-form Approach

Grether (1980 QJE)... Mobius, Niederle, Niehaus, and Rosenblat (2022 MS), Augenblick, Lazarus, Thaler (2025 QJE)

Two states, (*R*)ed and (*B*)lue, and a signal *s*

Bayes' rule:

$$p(R \mid s) = \frac{p(s \mid R)p(R)}{p(s)}, \quad p(B \mid s) = \frac{p(s \mid B)p(B)}{p(s)}$$

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Take the ratio (*p(s)* cancels):

$$\begin{aligned} \frac{p(R | s)}{p(B | s)} &= \frac{p(R)}{p(B)} \cdot \frac{p(s | R)}{p(s | B)} \\ \frac{p_t}{1 - p_t} &= \frac{p_{t-1}}{1 - p_{t-1}} \lambda_{s_t} \end{aligned}$$

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- Bayes' rule written in log form:

$$\log \left(\frac{p_t}{1 - p_t} \right) = \log \left(\frac{p_{t-1}}{1 - p_{t-1}} \right) + \log(\lambda_{s_t})$$

- p_t is the posterior, p_{t-1} is the prior, λ_{s_t} is the likelihood ratio of observing s_t

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- Empirical model:

$$y_t = \delta y_{t-1} + \beta \lambda_t + u_t$$

- $y_t = \log \left(\frac{p_t}{1 - p_t} \right)$, $\lambda_t = \log(\lambda_{s_t})$
- $\delta = \beta = 1$, $u_t = 0$ is Bayesian updating

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- Common Results: $\delta < 1$ (base rate neglect), $\beta < 1$ (conservatism)

Time Series Considerations: Classical Measurement Error

Basic Model: $y_t = \delta y_{t-1} + \beta \lambda + u_t$

Generalization:

(1) $y_t^* = \delta y_{t-1}^* + \beta \lambda_t + \epsilon_t$ (*unobserved; implementation error $\epsilon \sim N(0, \sigma_\epsilon)$*)

(2) $y_t = y_t^* + \eta_t$ (*observed; measurement error $\eta \sim N(0, \sigma_\eta)$*)

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For the OLS estimator of δ to be unbiased $\text{cov}(y_{t-1}, u_t) = 0$, however:

$$\text{cov}(y_{t-1}, u_t) = \text{cov}(y_{t-1}^* + \eta_{t-1}, \eta_t - \delta \eta_{t-1} + \epsilon_t) = -\delta V(\eta_{t-1})$$

Classical measurement error and OLS

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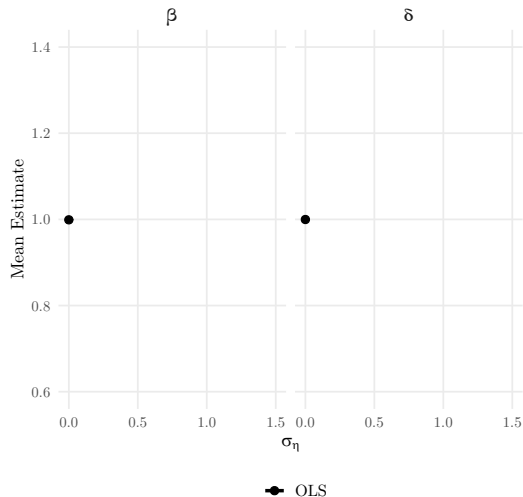
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This is a known issue in the literature, and (most) applications use an IV to avoid this problem

Simulations to Demonstrate the Impact of Classical Measurement Error

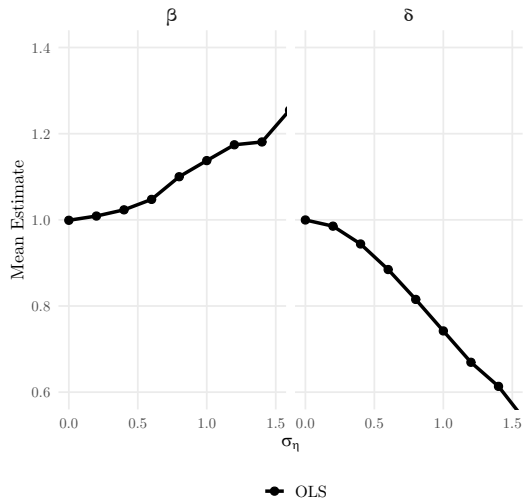
Simulate data for $\beta = \delta = 1$, $\sigma_\epsilon = 0.1$, and various σ_η

- Estimate β, δ using OLS



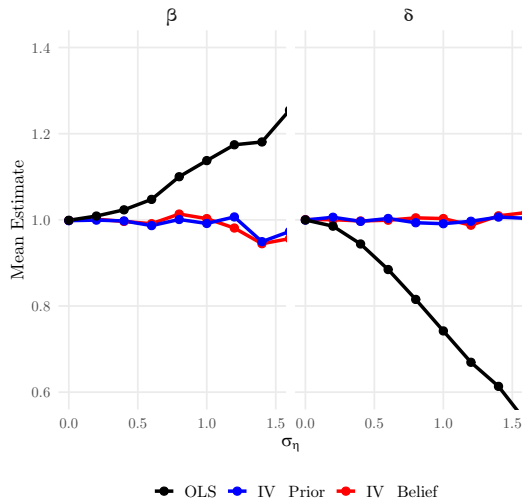
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Simulate data for $\beta = \delta = 1$, $\sigma_\epsilon = 0.1$, and various σ_η

- ▶ Estimate β , δ using OLS
- ▶ Solution: instruments
 - ▶ Mobius et. al. (2022): **prior**
 - ▶ Barron (2021): **lag-1 Bayesian beliefs**
- ▶ Takeaway: instruments solve the endogeneity problem associated with classical measurement errors
- ▶ **Problem: rounding is not a classical measurement error**



Aside: Broader Literature on Rounding Bias

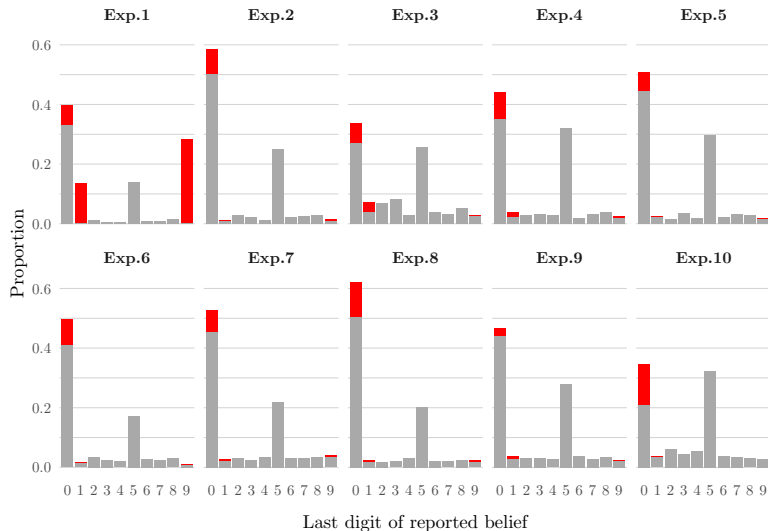
- ▶ Rosch (1975): round numbers are “cognitive reference points”
- ▶ Pope and Simonsohn (2010): round numbers as goals
- ▶ Ruud, Schunk, and Winter (2013): uncertainty causes rounding
- ▶ Manski and Molinari (2010), Giustinelli, Manski, Molinari (2022) rounding of probabilistic expectations in surveys
- ▶ To the best of our knowledge, rounding has not been studied in the context of belief updating
- ▶ To the best of our knowledge, no other application with rounding on both LHS *and* RHS of equation

(Mini) Meta Study

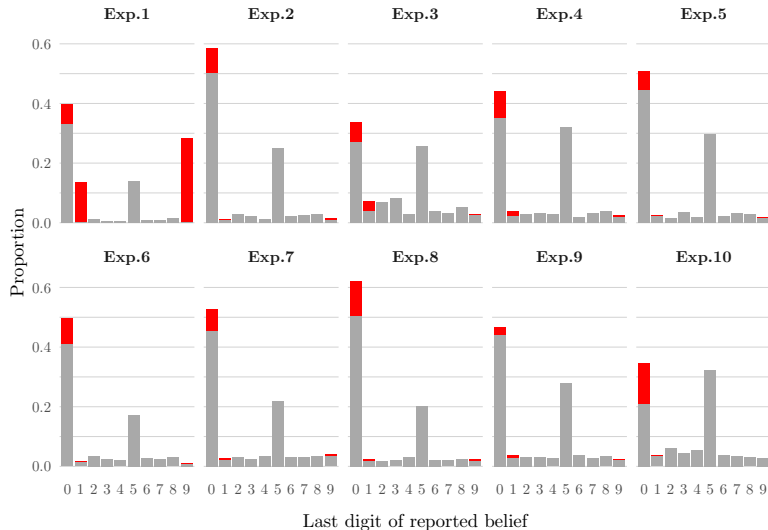
Recent Studies with Belief Updating and Direct Elicitation

Paper	Label	Context	Elicit.	Subj.	Signals	Scenarios	Approach	Boundary Resp.
<i>Hill (2017)</i>	Exp.1	Polit. facts	BDM	395	5	2	WLS	Replace
<i>Buser et al. (2018)</i>	Exp.2	Perform.	BDM	297	7	3	OLS	Drop
<i>Coutts (2019)</i>	Exp.3	Objective	BDM	318	4	2	OLS	Drop [†]
	Exp.4	Perform.	BDM	318	4	1		
	Exp.5	Weather	BDM	326	4	1		
<i>Barron (2021)</i>	Exp.6	Objective	QSR	222	6	5	IV	Replace
<i>Danz et al. (2022)</i>	Exp.7	Objective	BSR	299	3	10	N/A	N/A
	Exp.8	Objective	QSR	120	3	10		
<i>Möbius et al. (2022)</i>	Exp.9	Perform.	BDM	1057	4	1	IV	Drop [†]
<i>Oprea & Yuksel (2022)</i>	Exp.10	Objective	BSR	220	2	2	OLS	Replace

Data: Last digit of responses (excluding prior elicitation)

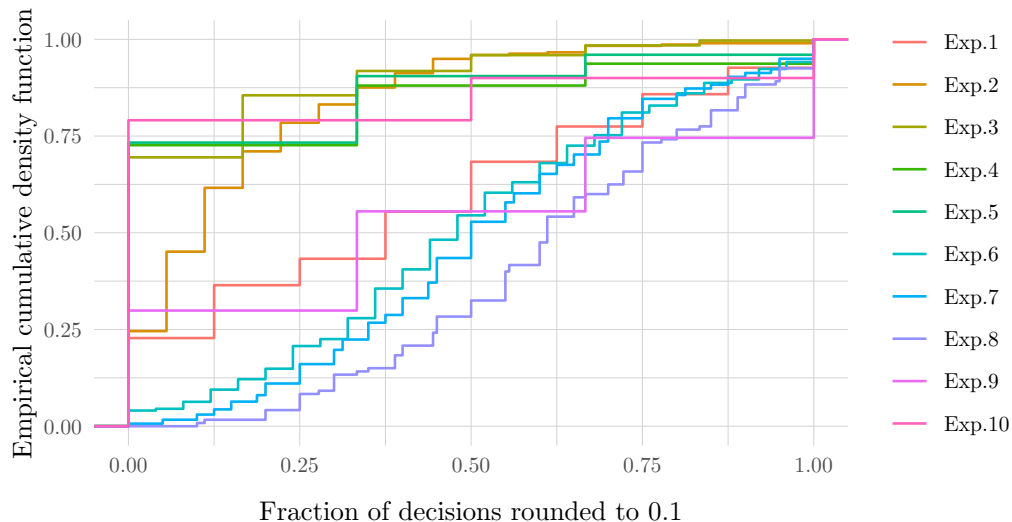


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Stylized Fact 1: Rounding of responses to belief elicitation tasks

Rounding is heterogeneous



Parameter Estimates for Frequent vs. Infrequent Rounders

	Estimates of δ				Estimates of β			
	Frequent rounders		Infrequent rounders	<i>p</i> -value	Frequent rounders		Infrequent rounders	<i>p</i> -value
Exp.1	0.876 (0.051)	$\sim$	0.835 (0.055)	0.584	0.642 (0.073)	$<$	0.909 (0.105)	0.036

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Exp.5	1.120 (0.042)	$\sim$	1.028 (0.048)	0.072	0.916 (0.062)	$>$	0.718 (0.064)	0.005
Exp.4	1.159 (0.053)	$\sim$	1.068 (0.042)	0.106	0.772 (0.064)	$>$	0.561 (0.055)	0.003
Exp.6	0.977 (0.021)	$\sim$	0.982 (0.022)	0.872	1.043 (0.080)	$>$	0.692 (0.067)	0.001
Exp.7	1.009 (0.056)	$\sim$	0.996 (0.031)	0.881	0.915 (0.057)	$>$	0.806 (0.037)	0.024
Exp.8	1.157 (0.054)	$\sim$	1.145 (0.069)	0.277	0.579 (0.052)	$>$	0.482 (0.052)	0.037
Exp.9	1.167 (0.045)	$>$	0.980 (0.028)	0.000	0.273 (0.017)	$>$	0.178 (0.022)	0.001
Exp.10	0.368 (0.128)	$\sim$	0.472 (0.075)	0.482	1.729 (0.137)	$>$	1.123 (0.062)	0.000

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Stylized Fact 2: Systematic differences between freq. and infreq. rounders

Potential Explanations

- ▶ Rounders may exhibit different behavioral patterns
- ▶ Rounding may cause problems for the estimation approach
- ▶ Both of the above
- ▶ Simulations + Experiment + New Approach for Estimation

Simulations

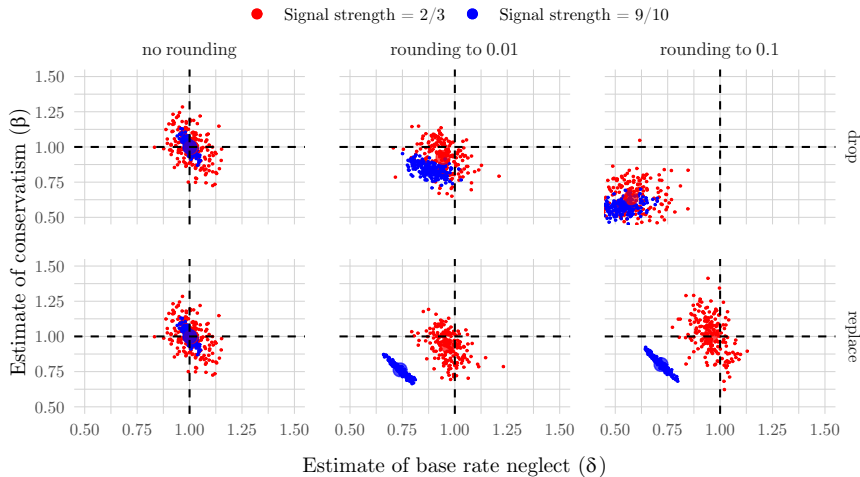
Simulation setup

Simulate data for a common experimental & econometric setup

- ▶ 50/50 prior
- ▶ Signal strength either $2/3$ or $9/10$
- ▶ 6 decisions per scenario
- ▶ IV using lagged Bayesian belief as instrument, either:
 - ▶ drop boundary decisions, or
 - ▶ replace them with 0.01/0.99

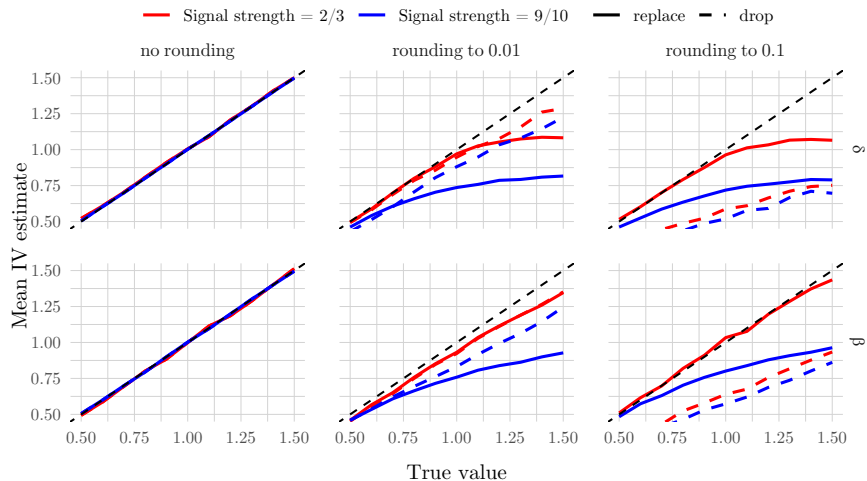
Simulated Experiments

Each experiment consisted of 100 Bayesian agents ($\beta = \delta = 1$, $\sigma_\epsilon = \sigma_\eta = 1$) w/ 50/50 prior



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Sensitivity of IV estimator

- ▶ When there is rounding, IV estimates of δ and β are sensitive to:
 - ▶ Prior (e.g., probability of red urn)
 - ▶ Signal strength (e.g., urn composition)
 - ▶ Signal realizations (e.g., sequence and number of draws)
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 - ▶ Stronger signal
 - ▶ Prior further away from 50% (not shown here)
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- ▶ This sensitivity is not present when there is no rounding

Experiment

Experimental Design

- ▶ Main Objective: Demonstrate the impact of rounding bias
- ▶ Approach: Present scenarios with various degrees of biases
 - ▶ For each subject, randomly generate 12 scenarios (i.e., vary prior, composition)
 - ▶ 4 decisions: after 0 signals, 9 signals, and two randomly picked from $\{1, \dots, 8\}$

Hypothesis: IV estimates predictably respond to the scenario selection

such that using scenarios with higher simulated estimates from forced-rounded Bayesian responses will lead to higher estimates of belief updating parameters in human subjects

Experimental Design

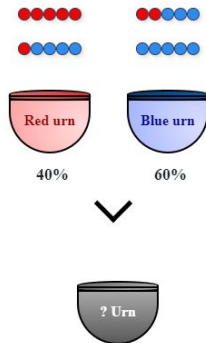
For Scenario 1, Red urn contains 6 red and 4 blue balls and a Blue urn contains 2 red and 8 blue balls. There is a 40 percent chance that the computer selects the Red urn, and 60 percent chance that the computer selects the Blue urn. The scenario is summarized to the right.

The selected urn is labeled '? Urn' and is displayed in gray. The selected urn is now fixed and stays the same for the entire Scenario 1. Your task will be to provide your best guess that the computer has selected the Red urn for Scenario 1. Reminder, the payment rule is designed so that **you can secure the largest chance of winning the prize by reporting your most-accurate guess**. The precise payment rule details are available at the end of the experiment.

Please enter **Your Second Guess**:

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Chance that the Red urn was selected: ?%
Chance that the Blue urn was selected: ?%

Submit Your Second Guess



History of Draws:



Experimental Design

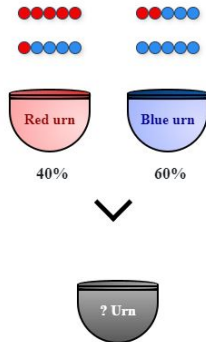
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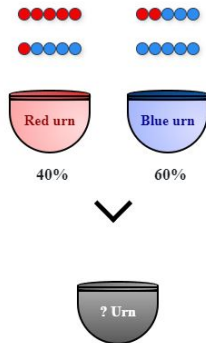


Design Remark 1: Context-neutral setting

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History of Draws:



Design Remark 2: Empty entry box on every decision

Experimental Design

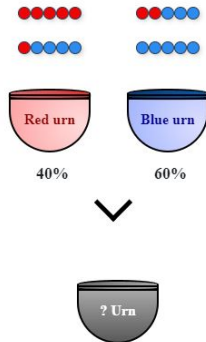
For Scenario 1, Red urn contains 6 red and 4 blue balls and a Blue urn contains 2 red and 8 blue balls. There is a 40 percent chance that the computer selects the Red urn, and 60 percent chance that the computer selects the Blue urn. The scenario is summarized to the right.

The selected urn is labeled '? Urn' and is displayed in gray. The selected urn is now fixed and stays the same for the entire Scenario 1. Your task will be to provide your best guess that the computer has selected the Red urn for Scenario 1. Reminder, the payment rule is designed so that **you can secure the largest chance of winning the prize by reporting your most-accurate guess**. The precise payment rule details are available at the end of the experiment.

Please enter **Your Second Guess**:

Your Second Guess:
Chance that the Red urn was selected: ?%
Chance that the Blue urn was selected: ?%

Submit Your Second Guess



History of Draws:



Experimental Design

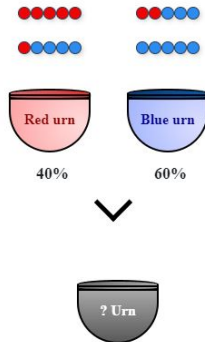
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History of Draws:



Design Remark 4: Urn composition need not be symmetric

Experiment Results

IV regression estimates

Procedure

1. Simulate Bayesian agents playing the experiment
2. Round beliefs to nearest 0.10
3. Estimate IV model

Do simulated IV estimates predict actual IV estimates?

IV regression estimates – pooled

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Role of simulations	-	scenarios sorted by estimates of δ			scenarios sorted by estimates of β		
Data	All	smallest	middle	largest	smallest	middle	largest
δ	0.932 (0.017)	0.830 (0.030)	0.939 (0.024)	1.023 (0.025)	0.960 (0.023)	0.967 (0.029)	0.813 (0.038)
β	0.460 (0.024)	0.467 (0.030)	0.534 (0.036)	0.429 (0.031)	0.363 (0.023)	0.582 (0.040)	0.664 (0.050)
Subjects	158	158	158	158	158	158	158
Scenarios Per Subject	12	4	4	4	4	4	4
Observations	5,673	1,893	1,875	1,896	1,893	1,875	1,896

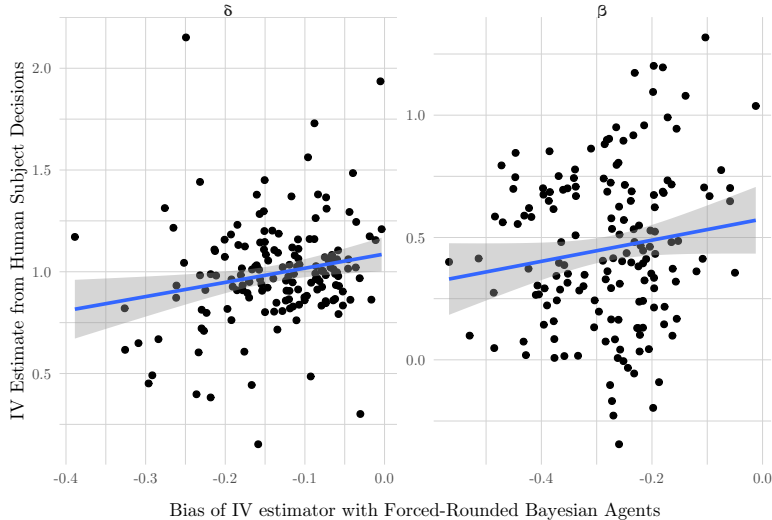
Result 1: Conservatism and base-rate neglect

IV regression estimates – pooled

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Result 2: Simulated estimates predict actual estimates – Hypothesis supported

IV Regression Estimates – individual



Result 2: Simulated estimates predict actual estimates – Hypothesis supported

Dealing with Rounding

A new estimator

Problem: IV estimator is not robust to rounding

Solution: Construct an estimator that:

- ▶ Treats rounded responses as *interval-valued*
- ▶ Still accounts for classical measurement error

Implementation of Interval-valued Approach

1. If a reported belief is an integer multiple of ten percentage points, then the actual belief was within five percentage points of the reported belief,
 - ▶ $0.30 \rightarrow (.250, .350)$
2. If a reported belief is an integer multiple of five percentage points (but not ten percentage points), then the actual belief was within 2.5 percentage points of the reported belief, and
 - ▶ $0.75 \rightarrow (.725, .775)$
3. If a reported belief is an integer multiple of one percentage point (but not a multiple of five or ten percentage points), then the actual belief was within half a percentage point of the reported belief.
 - ▶ $0.67 \rightarrow (.665, .675)$
4. Anything else is assumed to be within 0.0001

Bayesian implementation

Latent (logit) beliefs with measurement error:

$$y_t = \delta y_{t-1} + \beta \lambda_t + \eta_t - \delta \eta_{t-1} + \epsilon_t$$

y_0 = objective prior (assumed)

$$\epsilon_t \sim N(0, \sigma_\epsilon^2)$$

$$\eta_t \sim N(0, \sigma_\eta^2)$$

Data are interval-valued: $y_t \in (\underline{y}_t, \bar{y}_t)$

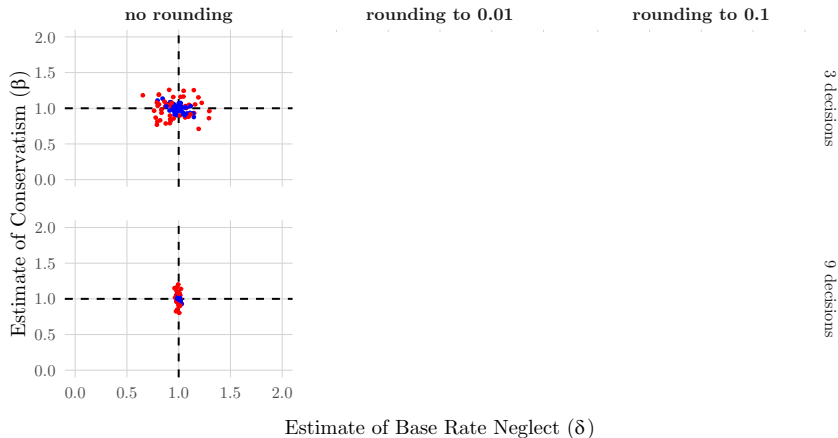
Implemented in *Stan*

I will discuss priors in a few slides

Simulation Experiments – interval model

Each experiment consisted of 100 Bayesian agents ($\beta = \delta = 1$, $\sigma_\epsilon = \sigma_\eta = 1$) facing 50/50 prior

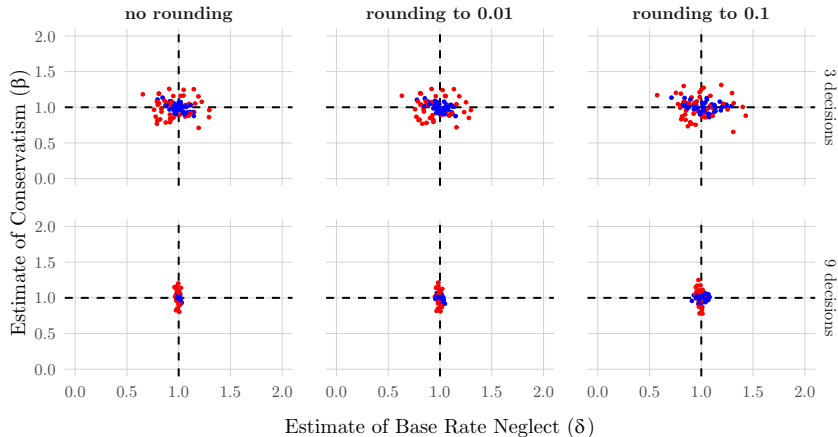
● signal strength = $2/3$ ● signal strength = $9/10$



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Sim. Result 4: proposed approach does well in recovering true parameters

But participants are heterogeneous!

Up to this point: pooled estimations assuming all participants have the same parameter values

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Problem: pooled models can suffer from *heterogeneity bias*

- ▶ Especially a problem for time-series models. e.g. Wilcox (2006)
- ▶ May also be a problem for non-linear models, but fortunately everything is linear here

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Cannot interpret the pooled model as estimating the mean of the heterogeneous parameter

Heterogeneity bias in time series models

For the moment, assume no classical measurement error or rounding

$$\begin{aligned} y_{i,t} &= \delta_i y_{i,t-1} + \beta_i \lambda_{i,t} + u_{i,t} \\ &= \bar{\delta} y_{i,t-1} + \bar{\beta} \lambda_{i,t} + \underbrace{u_{i,t} + (\delta_i - \bar{\delta}) y_{i,t-1} + (\beta_i - \bar{\beta}) \lambda_{i,t}}_{\text{econometric error term}} \end{aligned}$$

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Solution: model the heterogeneity. We use a hierarchical model

The hierarchical specification

Latent (logit) beliefs with measurement error (as before, but with i subscripts):

$$y_{i,t} = \delta_i y_{i,t-1} + \beta_i \lambda_{i,t} + \eta_{i,t} - \delta \eta_{i,t-1} + \epsilon_{i,t}$$
$$\epsilon_{i,t} \sim N(0, \sigma_{\epsilon,i}^2), \quad \eta_{i,t} \sim N(0, \sigma_{\eta,i}^2)$$

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Hierarchical structure (the new part):

$$\begin{pmatrix} \delta_i & \beta_i & \log \sigma_{\epsilon,i} & \log \sigma_{\eta,i} \end{pmatrix}^\top \sim \text{MultivariateNormal}(\mu, \Sigma)$$

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For the variance-covariance matrix, decompose into:

$$\begin{aligned}\Sigma &= \text{diag_matrix}(\tau)\Omega\text{diag_matrix}(\tau) \\ \tau_\delta, \tau_\beta, \tau_{\sigma_\epsilon}, \tau_{\sigma_\eta} &\sim \text{Cauchy}^+(0, 1) \\ \Omega &\sim \text{LKJ}(4)\end{aligned}$$

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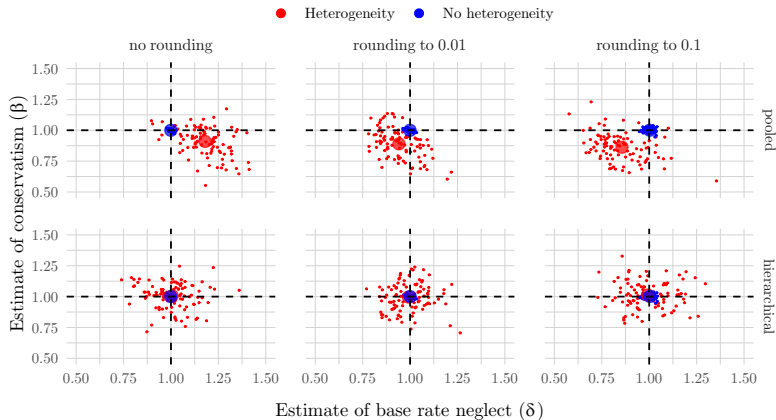
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We will show that a model with these hyper-priors recovers its parameters well

Simulation Experiments – Bayesian agents

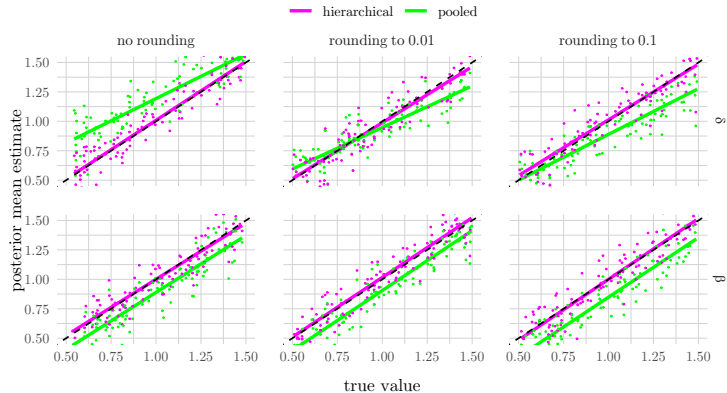
Each experiment consisted 20 “on average” Bayesian agents facing 50/50 prior. Each agent makes 4 decisions in 12 scenarios. “Heterogeneity” condition is $\delta_i, \beta_i \sim N(1, 0.25^2)$



Simulation Experiments – non-Bayesian agents

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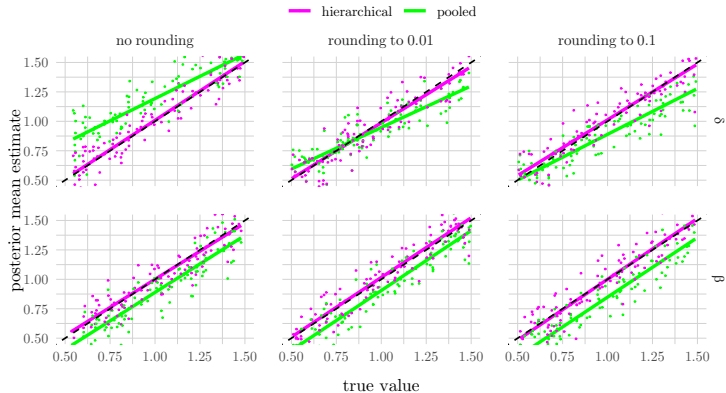
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We need the hierarchical model if heterogeneity is present

Model estimates – summary

Estimates	Pooled			Hierarchical
Parameter	OLS	IV	Interval	Interval
δ	0.773 (0.022)	0.932 (0.017)	0.973 (0.010)	0.969 (0.013)
β	0.502 (0.024)	0.460 (0.024)	0.468 (0.010)	0.642 (0.041)
Subjects	158	158	158	158
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Less base-rate neglect on average when using the hierarchical model

Hierarchical Model – population-level estimates

parameter	δ	β	$\log \sigma_\epsilon$	$\log \sigma_\eta$
μ	0.969 (0.013)	0.642 (0.041)	-0.463 (0.061)	-0.772 (0.096)
τ	0.121 (0.014)	0.481 (0.033)	0.670 (0.050)	0.923 (0.076)
Ω				
δ	1	0.510 (0.088)	0.131 (0.111)	-0.102 (0.112)
β	0.510 (0.088)	1	0.477 (0.078)	0.437 (0.077)
$\log \sigma_\epsilon$	0.131 (0.111)	0.477 (0.078)	1	0.439 (0.121)
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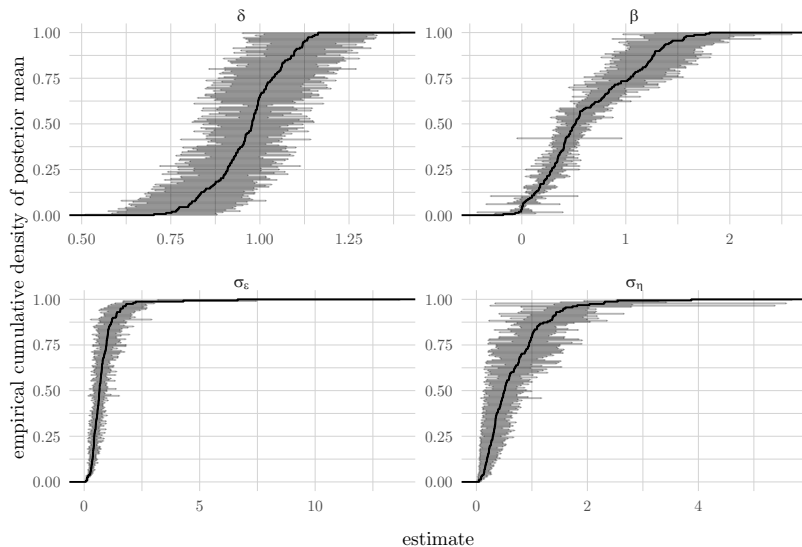
- Considerable heterogeneity between participants, especially in β

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- ▶ Considerable heterogeneity between participants, especially in β
- ▶ Substantial correlation between some parameters

Hierarchical model – individual-level estimates



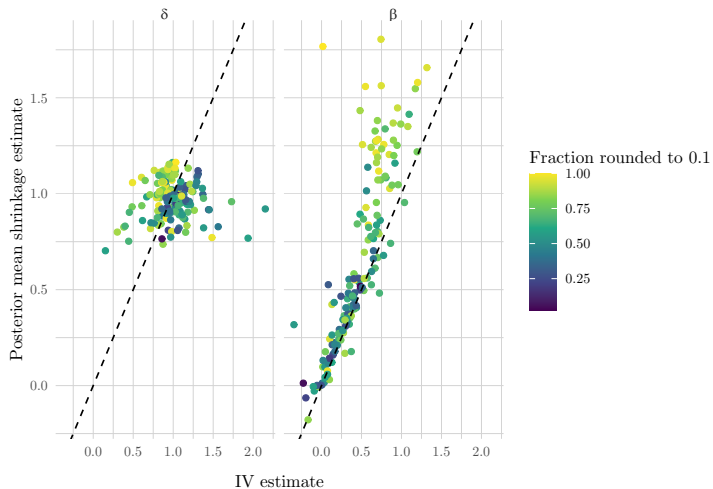
Classifying participants

Based on 95% Bayesian credible regions

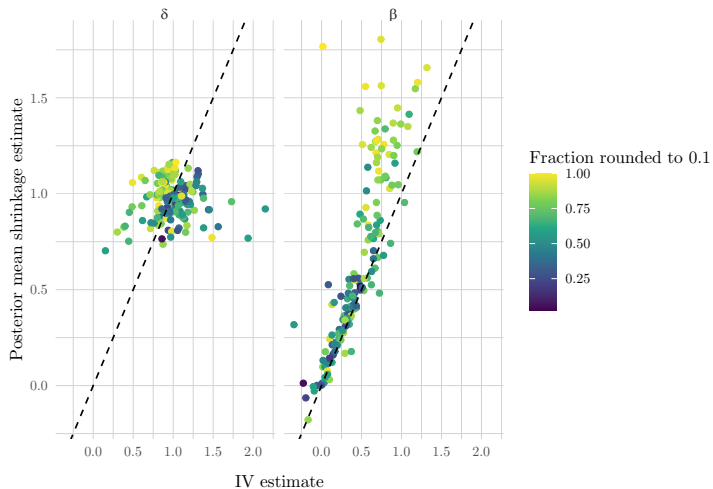
	fraction		
	< 1	contains 1	> 1
δ	0.177	0.778	0.044
β	0.658	0.266	0.076

- ▶ Majority of participants do not exhibit base-rate neglect ($\delta < 1$)
- ▶ Most participants are conservative ($\beta < 1$)

Comparison to individual IV estimates

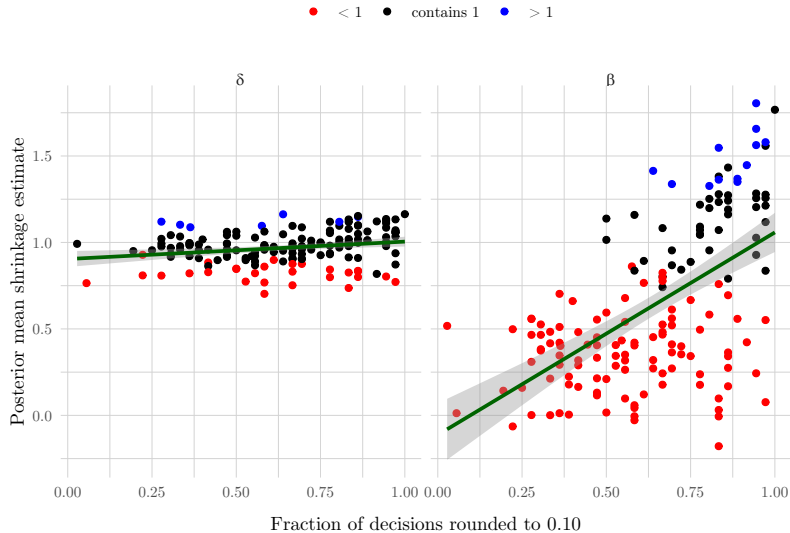


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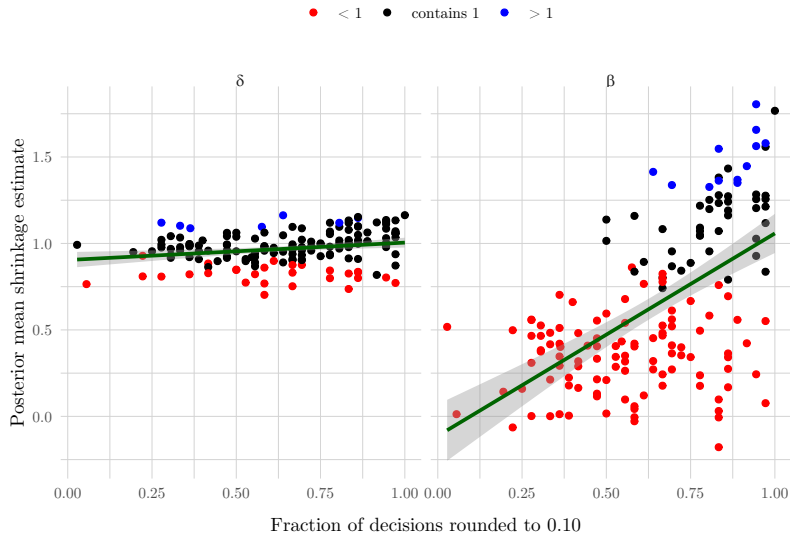


IV estimator does not do well for large β

Individual estimates and proclivity to round



Individual estimates and proclivity to round



There still exists a relationship between estimates and rounding

Determinants of updating behavior

Modify the hierarchical model to allow the means (μ) to depend on covariates:

- ▶ Number of correct answers to 10 Raven matrix questions
- ▶ Age (> 20 or ≤ 20)
- ▶ Female/Male
- ▶ STEM major
- ▶ Completed high school in US
- ▶ Fraction of decisions rounded to 0.1 (excluding first decisions)

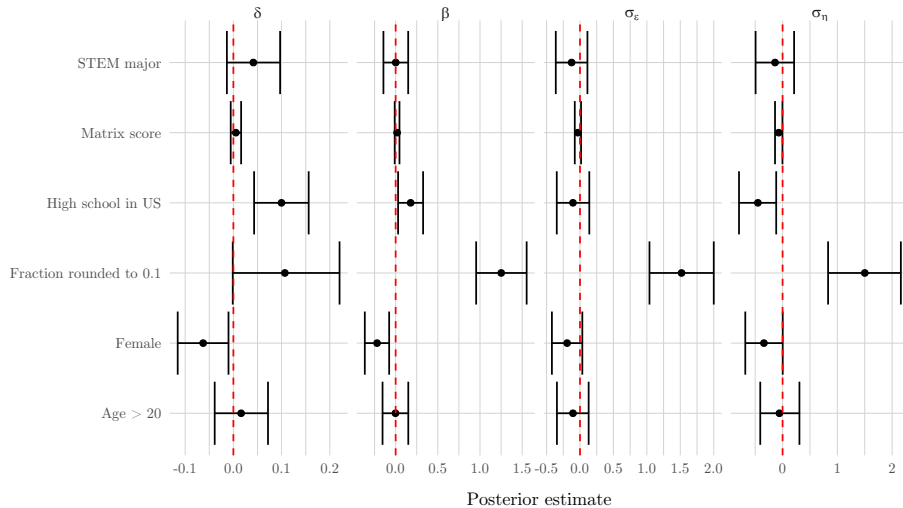
Removed participants who stated “prefer not to say” on any of these
Priors on all coefficients are $N(0, 1^2)$, covariates are de-meanned

Hierarchical model with covariates

parameter	δ	β	$\log \sigma_\epsilon$	$\log \sigma_\eta$
μ	0.966 (0.013)	0.653 (0.035)	-0.460 (0.055)	-0.772 (0.092)
τ	0.109 (0.013)	0.385 (0.028)	0.561 (0.049)	0.809 (0.078)
Matrix score	0.005 (0.006)	0.017 (0.015)	-0.030 (0.024)	-0.070 (0.035)
Age > 20	0.016 (0.028)	-0.002 (0.078)	-0.105 (0.121)	-0.058 (0.182)
Female	-0.063 (0.027)	-0.220 (0.073)	-0.193 (0.116)	-0.341 (0.172)
STEM major	0.041 (0.028)	0.001 (0.075)	-0.125 (0.121)	-0.138 (0.180)
High school in US	0.100 (0.029)	0.178 (0.074)	-0.105 (0.124)	-0.451 (0.173)
Fraction rounded to 0.1	0.107 (0.057)	1.252 (0.152)	1.519 (0.244)	1.501 (0.341)
Ω				
δ	1	0.459 (0.104)	0.068 (0.127)	-0.107 (0.123)
β	0.459 (0.104)	1	0.307 (0.098)	0.322 (0.091)
$\log \sigma_\epsilon$	0.068 (0.127)	0.307 (0.098)	1	0.230 (0.147)
$\log \sigma_\eta$	-0.107 (0.123)	0.322 (0.091)	0.230 (0.147)	1

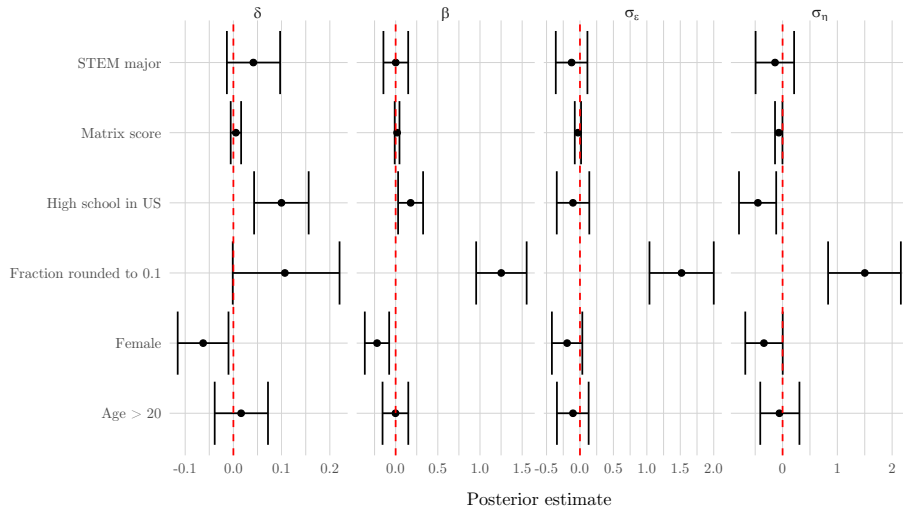
Determinants of δ and β

Posterior means (dots) and 95% credible regions (2.5th-97.5th percentile)



Determinants of δ and β

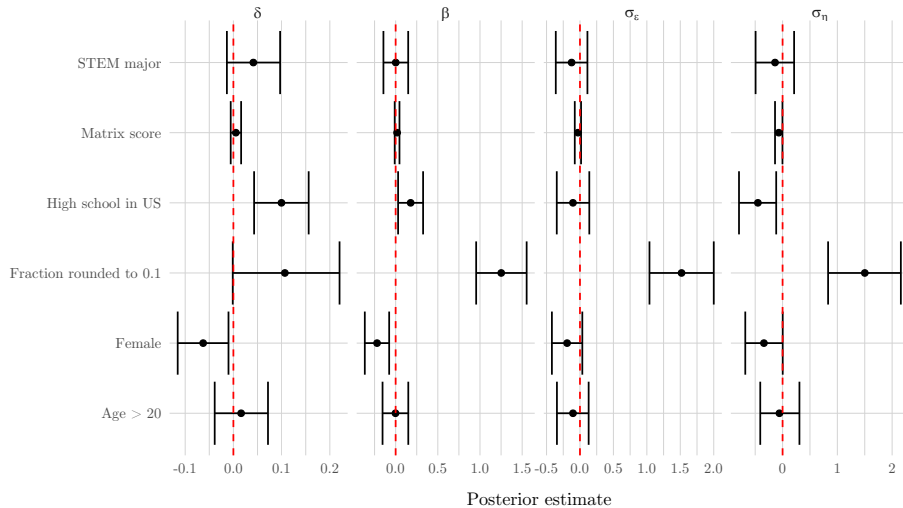
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Remark 1: High school in US – greater β, δ , less classical measurement error

Determinants of δ and β

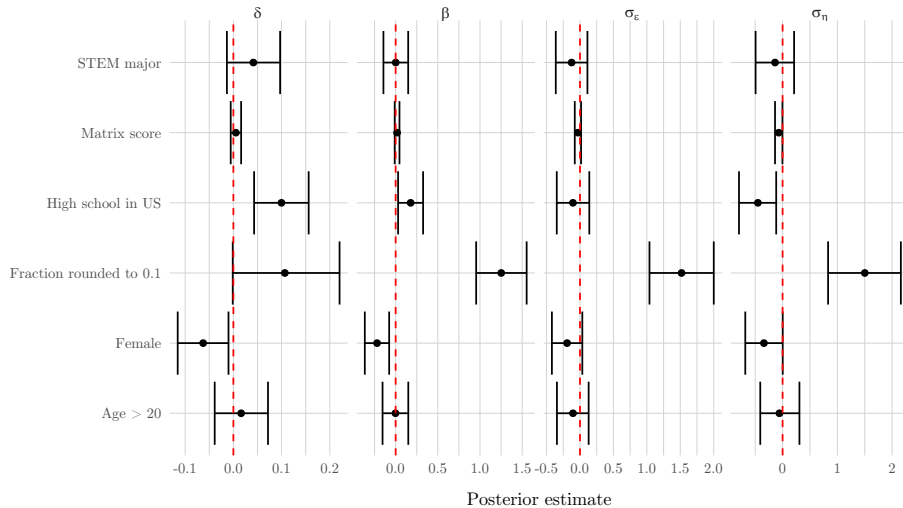
Posterior means (dots) and 95% credible regions (2.5th-97.5th percentile)



Remark 2: Fraction rounded – greater β , σ_ϵ , σ_η

Determinants of δ and β

Posterior means (dots) and 95% credible regions (2.5th-97.5th percentile)



Remark 3: Female – lower δ and β

Conclusion

- ▶ A simple model with economically relevant parameters: $y_t = \delta y_{t-1} + \beta \lambda_t + u_t$
 - ▶ IV leads to biased estimates because rounding and heterogeneity are prevalent

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 - ▶ Heterogeneity – estimate a hierarchical model
 - ▶ Example code at <https://github.com/JamesBlandEcon/RoundingNonBayesianCurveExample>

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- ▶ Relevant to *any* experiment where we elicit a number between 0 and 100
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 - ▶ Often our experiment's program will *force* responses to be rounded to one
- ▶ Key findings
 - ▶ Conservatism ($\beta < 1$) on average, but less than IV suggests
 - ▶ Much heterogeneity in estimated parameters

Using our estimator

<https://github.com/JamesBlandEcon/RoundingNonBayesianCurveExample>

- ▶ Runs in *R* using just the *RStan* library
- ▶ Example code for simulating data, estimating the model, and analyzing results
- ▶ Pooled and hierarchical specifications

Currently available:

- ▶ Objective priors
- ▶ Symmetric treatment of signals

Working on:

- ▶ Subjective priors
 - ▶ Prior is *estimated*
 - ▶ Validated with objective priors – works well
- ▶ “Good news / bad news” distinction for motivated beliefs

Thank You!

Questions? Comments?